Lexiao Lai

Contact

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Employment

The University of Hong Kong — Hong Kong, China
Assistant Professor, Department of Mathematics — July 2024–

Research Interests

Nonconvex optimization, applied semialgebraic geometry, data science

Education

Columbia University in the City of New York — New York, U.S.
Ph.D. in Operations Research [thesis] — Sept. 2019–May 2024
Advisor: Cédric Josz

The University of Hong Kong — Hong Kong, China
B.Sc. in Mathematics — Sept. 2015–June 2019

Publications

1. Phase transitions in phase-only compressed sensing (with Junren Chen and Arian Maleki), *IEEE International Symposium on Information Theory (ISIT)*, 2025 [preprint] [proceeding]
2. Nonsmooth rank-one symmetric matrix factorization landscape (with Cédric Josz), *Optimization Letters*, 2025 [preprint] [journal]
3. Sufficient conditions for instability of the subgradient method with constant step size (with Cédric Josz), *SIAM Journal on Optimization*, 2024 [preprint] [journal]
4. Convergence of the momentum method for semialgebraic functions with locally Lipschitz gradients (with Cédric Josz and Xiaopeng Li), *SIAM Journal on Optimization*, 2023 [preprint] [journal]
5. Global stability of first-order methods for coercive tame functions (with Cédric Josz), *Mathematical Programming*, 2023 [preprint] [journal]
6. Lyapunov stability of the subgradient method with constant step size (with Cédric Josz), *Mathematical Programming*, 2023 [preprint] [journal]
7. Nonsmooth rank-one matrix factorization landscape (with Cédric Josz), *Optimization Letters*, 2022 [preprint] [journal]
8. Time-dependent surveillance-evasion games (with Elliot Cartee, Qianli Song, and Alexander Vladimirsky), *IEEE Conference on Decision and Control (CDC)*, 2019 [preprint] [proceeding]

Preprints

1. Convergence of difference inclusions via a diameter criterion (with Mingzhi Song), 2026 [preprint]
2. Manifold constrained steepest descent (with Kaiwei Yang), 2026 [preprint]
3. Certifying optimality in nonconvex robust PCA (with Pinxi Gong and Jianhao Ma), 2026 [preprint]
4. Global convergence of the subgradient method for robust signal recovery (with Zesheng Cai and Tiansheng Li), 2026 [preprint]
5. Non-convex self-concordant functions: Practical algorithms and complexity analysis (with Donald Goldfarb, Tianyi Lin, and Jiayu Zhang), 2025 [preprint]
6. On the diameter of subgradient sequences in o-minimal structures (with Mingzhi Song), 2025 [preprint]
7. Proximal random reshuffling under local Lipschitz continuity (with Cédric Josz and Xiaopeng Li), 2024 [preprint]

Talks

1. Hong Kong PolyU Applied Mathematics Seminar, Hong Kong, June 17th 2026, *On the diameter of subgradient sequences in o-minimal structures*
2. SIAM Conference on Optimization, Edinburgh, June 5th 2026, *On the diameter of subgradient sequences in o-minimal structures*

3. HKU Mathematics for Machine Learning Seminar, Hong Kong, April 16th 2026, *Certifying optimality in nonconvex robust PCA*
4. International Conference on Scientific Machine Learning, Hong Kong, December 3rd 2025, *On the diameter of subgradient sequences in o-minimal structures*
5. International Conference on Scientific Machine Learning, Hong Kong, December 3rd 2024, *Proximal random reshuffling under local Lipschitz continuity*
6. 14th Triennial International Conference of APORS, Hangzhou, November 17th 2024, *Proximal random reshuffling under local Lipschitz continuity*
7. International Symposium on Mathematical Programming, Montréal, July 23rd 2024, *Global stability of first-order methods for coercive tame functions*
8. IMS Young Mathematical Scientists Forum – Applied Mathematics, Singapore, January 9th 2024, *Global stability of first-order methods for coercive tame functions*
9. INFORMS Annual Meeting, Phoenix, October 17th 2023, *Global stability of first-order methods for coercive tame functions*
10. UCSD Optimization and Data Science Seminar, San Diego, October 4th 2023, *Global stability of first-order methods for coercive tame functions*
11. International Congress on Industrial and Applied Mathematics, Tokyo, August 24th 2023, *Global stability of first-order methods for coercive tame functions*
12. SIAM Conference on Optimization, Seattle, June 1st 2023, *Global stability of first-order methods with constant step size for coercive tame functions*
13. CUHK SEEM Department Seminar, Hong Kong, December 8th 2022, *Lyapunov stability of the subgradient method with constant step size*
14. HKU Optimization and Machine Learning Seminar, Hong Kong, December 6th 2022, *Lyapunov stability of the subgradient method with constant step size*
15. PGMODAYS, Paris, November 29th 2022, *Lyapunov stability of the subgradient method with constant step size*
16. INFORMS Annual Meeting, Indianapolis, October 17th 2022, *Lyapunov stability of the subgradient method with constant step size*

Grants

1. ECS grant 27301425, *A study of first-order methods for structured nonconvex optimization problems in data science*, HKD778,276, Jan. 2026–Dec. 2028

Awards & Honours

- Silver Reviewer Award, ICML — 2026
- Columbia IEOR Department Fellowship — 2019
- Walter Brown Memorial Prizes in Mathematics, HKU — 2019
- Doris Chen Undergraduate Project Prize, HKU — 2018
- Liu Ming-Chit Prize in Mathematics, HKU — 2018
- Alan John Allis Prize in Mathematics, HKU — 2016, 2017
- Dean's Honours List, HKU — 2016, 2017, 2019
- HKSAR Government Scholarship, HKU — 2015–2019

Teaching Experience

Course instructor

HKU: Operations Research I — 2025–

Teaching assistant

Columbia:

- Optimization Methods & Models — Spring 2024
- Optimization Methods & Models for Financial Engineering — Fall 2023
- Convex Optimization — Spring 2023

HKU: Linear Algebra I — Spring 2019

Student Supervision

Ph.D. students

- Kaiwei Yang — 2025–
- Mingzhi Song — 2024–

Undergraduate students

- Zesheng Cai, Shiyang Chen, Yize Chu, Pinxi Gong, Tiansheng Li, Qi Peng, Shuaihua Wang, Han Zheng, Zhuofei Zhu

Service

Session chair

- *Advances in Nonsmooth, Nonconvex, and Stochastic Optimization*, SIAM Conference on Optimization, 2026
- *Optimization Theory and Algorithms I*, APORS Youth Forum, 2025
- *Recent advances in first-order methods*, International Conference on Continuous Optimization (ICCOPT), 2025
- HKMS-HKSIAM Joint Young Scholars Symposium, 2024
- *First-order methods and large-scale optimization*, International Symposium on Mathematical Programming (ISMP), 2024
- *Structured and tame optimization*, INFORMS Annual Meeting, 2023

Reviewer

- Annual Conference on Artificial Intelligence and Statistics (AISTATS)
- Applied Mathematics and Optimization
- Computational Optimization and Applications
- International Conference on Machine Learning (ICML)
- Journal of Optimization Theory and Applications
- Mathematics of Operations Research
- Set-Valued and Variational Analysis
- SIAM Journal on Optimization

Thesis defense committees

- Zimeng Wang, HKU
- Tan Zhang, HKU

Knowledge Exchange

- **TouchStart Science Summer Programme**, HKU Academy for the Talented July 2026
  The Mathematical Foundations Behind AI [website]: led project discussion and sessions on computer vision and linear algebra for 100 secondary students.

Internship

TCL Corporate Research (Hong Kong) Company Limited Hong Kong, China
Research Intern, AI Research Lab May–Sept. 2021

Computer Skills

Python, MATLAB, LaTeX